

LUKE PARRELLA

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SKILLS

Languages: Python, C, MATLAB, LaTeX

Technology: Microcontrollers, Radio Communication, Observability with Prometheus and Grafana

PROJECTS

BO-AT Autonomous sailing vessel

August 2025 - Present

- Work on a team of seven engineers to develop a system for carrying a modular payload autonomously throughout ocean waters
- Develop long range communication between control computer and vessel, ensuring all electronics fit safely within the hull
- Communicate with faculty mentor weekly to develop requirements that align the project with client objectives and ensure market viability

Captain Capacity - digital room occupancy monitor

February 2024 - May 2024

- Programmed logic for real time room occupancy monitoring, adhering to building safety protocols
- Developed a unique one-way automatic door locking system, allowing for occupants to leave without anyone being able to enter when room capacity is reached
- Delivered a final product pitch, demonstrating valuable presentation and marketing skills

WORK EXPERIENCE

Boston University Office of the Provost

Boston, MA

Student Worker

September 2025 - Present

- Maintain daily duties around the office to ensure smooth operations by full time employees
- Communicate with Provost staff and school deans to coordinate tasks and run events

HITA Energy

Boston, MA

Founding Engineer

October 2024 - June 2025

- Founding engineer at an angel backed startup, where I worked on a cross disciplinary team including software specialists and material engineers
- Designed and implemented control system in python that interfaced with real hardware including pneumatic controls and bitcoin ASIC mining units
- Designed and built hardware control system containing relays to control high and low voltage power in order to command and receive data from pumps, valves, and thermistors

University of South Carolina

Remote

Researcher

June 2024 - July 2024

- Upgraded an existing MATLAB program and developed two new types of artificial microscopy images for DNN autoencoder training.
- Applied computational geometry to map 3D points onto 2D projections.
- Delivered a final comprehensive presentation covering the entire research project, highlighting the performance enhancements made to the autoencoder, and how it is improved by our new images.

EDUCATION

Boston University

Boston, MA

Bachelor of Science Computer Engineering

Expected Graduation May 2026

Relevant courses: Signals & Systems, Deep Learning, Computer Networking, Operating Systems